Test Series No.

1.9

MASTERY TEST SERIES

DURATION: 180 Minutes | **M. MARKS: 720**

DATE: 10/03/2026

SYLLABUS

Physics:	11th Full Syllabus
Chemistry:	11th Full Syllabus
Botany:	11th Full Syllabus
Zoology:	11th Full Syllabus

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration and the Test Booklet contains **180** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 720.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR INSTRUCTIONS:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS): _____

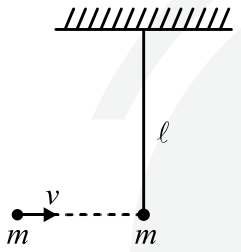
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OMR Bar Code Number: _____

Student's Signature: _____ **Invigilator's Signature:** _____

1. Two balls A and B having masses 2 kg and 4 kg respectively are placed on the y-axis. If ball A is at point (0, 5), then find the possible coordinates of ball B such that center of mass lies on the y-axis at a distance of 1 unit from the origin.
(1) (0, –1) and (0, –4)
(2) (0, –2) and (0, –4)
(3) (0, –1) and (0, –2)
(4) (0, 0) and (0, –4)

2. A simple pendulum has a bob of mass m which is suspended from a string of length ℓ. A ball having same mass hits the pendulum perfectly inelastically as shown. Find the minimum speed v of the first ball such that the system of ball and bob just complete the vertical circle. (g is acceleration due to gravity)



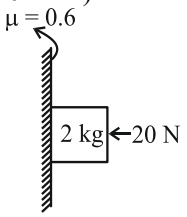
(1) $\sqrt{5g\ell}$ (2) $2\sqrt{5g\ell}$
(3) $3\sqrt{5g\ell}$ (4) $4\sqrt{5g\ell}$

3. Two projectiles are projected at complementary angles to the horizontal with same speed 20 m/s. One of them rises 10 m higher than the other. Find the angles of projection. (Take g = 10 m/s^2)
(1) 20° and 70°
(2) 37° and 53°
(3) 15° and 75°
(4) 30° and 60°

4. A policeman standing 100 m behind a thief on a straight line observes the thief running with constant speed 10 m/s. With what constant speed should the policeman start running, to catch the thief in 40 s?
(1) 20.0 m/s (2) 12.5 m/s
(3) 18.0 m/s (4) 15.0 m/s

5. In a particular system, the units of length, mass and time are chosen to be 10 cm, 10 g and 0.1s respectively. The unit of force in this system will be equal to
(1) 0.1 N (2) 1 N
(3) 10 N (4) 100 N

6. If a block of mass 2 kg is pushed normally against a rough wall with a force of 20 N as shown in the figure, the kinetic frictional force acting on the block is: (g = 10 m/s^2)



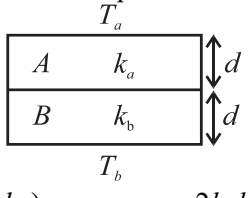
(1) 10 N (2) 12 N
(3) 20 N (4) Zero

7. A conical pendulum consists of a mass attached to a string of length $L = 5$ meters. The mass performs uniform circular motion such that the string makes a constant angle $\theta = 37°$ with the vertical. Calculate the time period of the pendulum. (Take acceleration due to gravity $g = 10$ m/s^2 and assume $\pi^2 = 10$).
(1) 2.0 s (2) 3.0 s
(3) 4.0 s (4) 5.0 s

8. If the differential equation of a particle performing SHM along X-axis is given by $0.5\frac{d^2x}{dt^2} + 4x = 0$, then the time period of SHM is; (here x is in m and t is in s)
(1) $\frac{\pi}{2}$ s (2) $\frac{\pi}{3}$ s
(3) π s (4) $\frac{\pi}{\sqrt{2}}$ s

9. If an ideal gas follows a process described by the equation P^2V = constant (where P is pressure and V is volume), from the initial state (P_1, V_1, T_1) to final state (P_2, V_2, T_2), then;
(1) If $P_1 > P_2$ then $T_1 > T_2$
(2) If $V_2 > V_1$ then $T_2 > T_1$
(3) If $V_2 > V_1$ then $T_2 < T_1$
(4) If $P_1 > P_2$ then $V_1 > V_2$

10. Two rods A and B of same geometry and different materials are welded together as shown in figure. Heat flow from top surface to bottom surface. Their thermal conductivities are k_a and k_b. The thermal conductivity of the composite rod will be:
(T_a and T_b are the temperature at outer faces of rod)



(1) $\frac{3(k_a + k_b)}{2}$ (2) $\frac{2k_ak_b}{k_a + k_b}$
(3) $2(k_a + k_b)$ (4) $\frac{k_a + k_b}{2}$

[Contd...

11. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: By increasing temperature, the peak emission wavelength of a black body shifts to lower wavelength.

Reason R: Peak emission wavelength of a black body is proportional to the fourth power of temperature.

In the light of the above statements, choose the **correct** answer from the options given below:

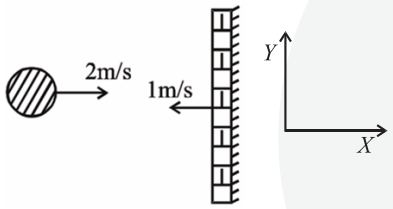
(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true but R is NOT the correct explanation of A.

12. A ball is moving with velocity 2 m/s towards a heavy wall moving towards the ball with speed 1 m/s as shown in figure. Assuming collision to be elastic, the velocity of ball immediately after collision is:



(1) $4\hat{i}$ m/s (2) $2\hat{i}$ m/s

(3) $-4\hat{i}$ m/s (4) $-2\hat{i}$ m/s

13. If a graph is plotted between T^2 and r^3 for a satellite moving in circular path around a planet then its slope will be: (T is the time period of satellite revolving around a planet of mass M in a orbit of radius r and G is universal gravitational constant)

(1) $\frac{4\pi^2}{GM}$ (2) $\frac{GM}{4\pi^2}$

(3) $4\pi GM$ (4) zero

14. A particle of mass m is driven by a machine that delivers a constant power of k watt. If the particle starts from rest the force on the particle at time t is:

(1) $\sqrt{mkt}$

(2) $\sqrt{\frac{2mk}{t}}$

(3) $\frac{1}{2}\sqrt{\frac{mk}{t}}$

(4) $\sqrt{\frac{mk}{2t}}$

15. Consider a spherical body falling through a viscous liquid of infinite extent, if it finally attains a constant velocity then:

(1) upthrust + weight = viscous drag

(2) weight + viscous drag = upthrust

(3) viscous drag + upthrust = weight

(4) viscous drag + upthrust > weight

16. A body of mass 0.5 kg is thrown vertically upwards with 20 m/s. If it again reaches ground with a speed of 16 m/s then find work done by air resistance in whole journey.

(1) – 60 J (2) –36 J

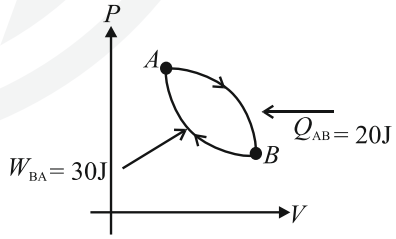
(3) –70 J (4) 60 J

17. In a guitar, two strings A and B made of same material are slightly out of tune and produce beats of frequency 6 Hz. When tension in B is slightly decreased, the beat frequency increases to 7 Hz. If the original frequency of B is 530 Hz, the frequency of A will be:

(1) 537 Hz (2) 523 Hz

(3) 524 Hz (4) 536 Hz

18. In a cyclic process shown in the figure an ideal gas is adiabatically taken from B and A, the work done on the gas during the process $B \rightarrow A$ is 30 J, when the gas is taken from $A \rightarrow B$ the heat absorbed by the gas is 20 J. The change in internal energy of the gas in the process $A \rightarrow B$ is:



(1) 20 J (2) – 30 J

(3) 50 J (4) – 10 J

19. A container having 6 mole of ideal gas at a temperature 30°C has movable piston which maintains constant pressure of 1 atm in container. Heat is supplied to the gas until temperature becomes 80°C. The work done by the gas is nearly: (take ideal gas constant = 8.3 J $mol^{-1}K^{-1}$)

(1) 5 Kilo Joule

(2) 2.5 Kilo Joule

(3) 7.5 Kilo Joule

(4) 3 Kilo Joule

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20. Pitch of the screw gauge is 0.5 mm and there are 100 divisions on its circular scale. When nothing is put in-between jaws, the zero of the circular scale lies 2 divisions below the reference line. While measuring the diameter of wire, it shows 8 divisions on main scale and 83th division of circular scale coincides with reference line. The diameter of wire will be:

(1) 2.105 mm

(2) 2.405 mm

(3) 4.405 mm

(4) 4.624 mm

21. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: A body always have zero acceleration if its velocity is zero at a given instant of time.

Reason R: A body is momentarily at rest when it reverses its direction of motion.

In the light of the above statements, choose the **correct** answer from the options given below:

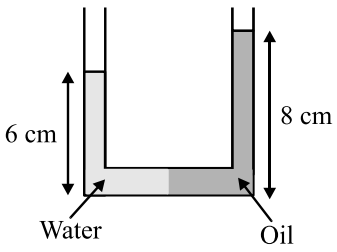
(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true but R is not the correct explanation of A.

22. Consider a U-shaped tube as shown in figure in which water and oil are put in its left side and right side respectively. If heights from the bottom for water and oil columns in it are 6 cm and 8 cm respectively then density of oil in tube is: [take $\rho_{water} = 10^3$ kg/m^3]



(1) 9×10^2 kg/m^3

(2) 7.5×10^2 kg/m^3

(3) 5×10^2 kg/m^3

(4) 3.5×10^2 kg/m^3

23. Suppose a particle starting from rest, moves in circular path of radius r meter with constant angular acceleration. If it attains speed equal to 6 m/s after completing its 3 rounds, then magnitude of its angular acceleration will be:

(1) $\frac{3}{r}$ rad/s^2 (2) $\frac{3}{\pi r}$ rad/s^2

(3) $\frac{3}{\pi r^2}$ rad/s^2 (4) $\frac{5}{2\pi r}$ rad/s^2

24. If a cylinder contains hydrogen gas at pressure of 498 kPa and temperature 127°C then density of hydrogen gas in container is: (R = 8.3 J mol^{-1} K^{-1})

(1) 0.3 kg/m^3 (2) 0.5 kg/m^3

(3) 0.2 kg/m^3 (4) 0.1 kg/m^3

25. If a uniform wire of length l and cross-sectional area A is made of a material of Young's modulus Y then work done in stretching the wire slowly by an amount x is given by:

(1) $\frac{YAx^2}{l}$ (2) $\frac{YAx^2}{2l}$

(3) $\frac{YAl^2}{x}$ (4) $\frac{YAl^2}{2x}$

26. If the escape velocity of the body which is thrown from earth's surface is v_e, then the escape velocity of the body if it is thrown at an angle θ with horizontal from earth's surface is:

(1) v_e (2) $v_e\cos\theta$

(3) $v_e\sin\theta$ (4) $v_e\tan\theta$

27. The stream of a river is flowing with a speed of 2 km/h. A swimmer can swim at a speed of 4 km/ h. What should be the direction of the swimmer with respect to the direction of river flow to cross the river reaching exactly the opposite end?

(1) 60º (2) 120º

(3) 90º (4) 150º

28. A standing wave is formed in a horizontal string of length 150 cm and mass 1.5 kg tied at both ends. Tension in the string is 36 N. There are 4 nodes on the string other than the ends. Calculate the natural frequency of the wave?

(1) 20 Hz (2) 10 Hz

(3) 30 Hz (4) 40 Hz

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29. Radius of gyration of a thin uniform rod of length l about an axis passing through its centre and perpendicular to its length is:

(1) $\frac{l}{2\sqrt{3}}$ (2) $0.5l$

(3) $\frac{l}{\sqrt{3}}$ (4) $\frac{l}{\sqrt{2}}$

30. 10 gm of ice at –20°C is dropped into a calorimeter containing 10 gm of water at 10°C. When equilibrium is reached, the calorimeter will contain (the specific heat of water is twice that of ice):

(1) 20 gm of water

(2) 20 gm of ice

(3) 10 gm ice and 10 gm of water

(4) 5 gm ice and 15 gm of water

31. A flask is filled with 4g of a gas at 12°C and then heated to 50°C. Owing to the escape of the gas, the pressure in the flask is the same at the beginning as well as at the end of the experiment. Then the mass of the gas that has escaped is nearly:

(1) 4 g (2) 3.53 g

(3) 0.47 g (4) 3.06 g

32. A person is standing in an elevator. In which situation he finds his apparent weight less than actual weight;

(1) the elevator accelerates upward with constant acceleration.

(2) the elevator accelerates downward with constant acceleration.

(3) the elevator moves upward with uniform velocity.

(4) the elevator moves downward with uniform velocity.

33. The displacement of a particle executing simple harmonic motion is given by;

$y = A_0 + A\sin\omega t + B\cos\omega t$

Then the amplitude of its oscillation is given by:

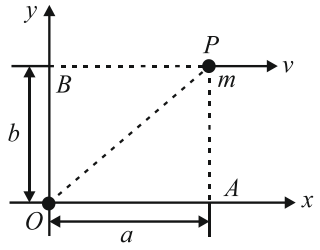
(1) $A_0 + \sqrt{A^2 + B^2}$

(2) $\sqrt{A^2 + B^2}$

(3) $\sqrt{A_0^2 + (A+B)^2}$

(4) $A + B$

34. A particle (P) of mass m is moving along a straight line parallel to x-axis with constant velocity v. Find angular momentum about the origin.



(1) $+mv^2b\hat{k}$ (2) $-mvb\hat{k}$

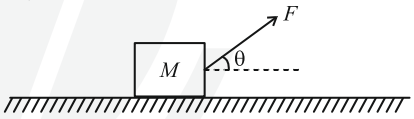
(3) $-2mvb\hat{k}$ (4) $-mva\hat{k}$

35. If infinite number of particles each of mass 2 kg are placed along x-axis at x = 2 m, 4 m, 8 m, 16 m, ……∞ then magnitude of net gravitational field (in N/kg) at origin is:
(G = universal gravitational constant)

(1) $2G$ (2) $\frac{G}{3}$

(3) $\frac{2G}{3}$ (4) ∞

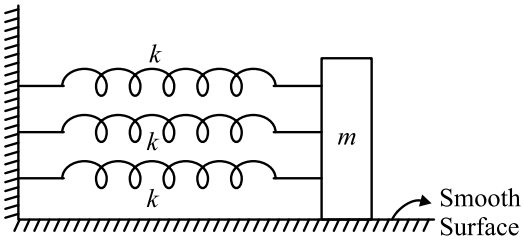
36. Suppose block is placed on a rough horizontal surface as shown. By varying the angle θ, minimum magnitude of force F required to slide the block is; (μ is coefficient of friction between block and ground)



(1) $\frac{2\mu Mg}{\sqrt{1+\mu^2}}$ (2) μMg

(3) $\frac{\mu Mg}{\sqrt{1+\mu^2}}$ (4) $\frac{\mu Mg}{\sqrt{1+2\mu^2}}$

37. Mass m is connected symmetrically with identical springs having spring constant k as shown in the figure. The time period of oscillation of mass m will be:



(1) $2\pi\sqrt{\frac{m}{k}}$ (2) $2\pi\sqrt{\frac{3m}{k}}$

(3) $2\pi\sqrt{\frac{m}{3k}}$ (4) $3\pi\sqrt{\frac{m}{k}}$

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38. The linear density of a vibrating string is 1.3×10^{-4} kg/m. A transverse wave is propagating on the string and is described by the equation $y = 0.2\sin(4.5x - 18t)$ where x and y are measured in meter and t is in second. The tension in the string is:

(1) 20.8×10^{-3} N (2) 20.8×10^{-4} N
(3) 20.8×10^{-2} N (4) 20.8×10^{-5} N

39. The root mean square velocity of hydrogen molecules at 300 K is 1930 m/s. The r.m.s. velocity of oxygen molecules at 1200 K will be:

(1) 482 m/s (2) 965 m/s
(3) 1930 m/s (4) 3860 m/s

40. What is the number of significant figures in 0.310×10^3?

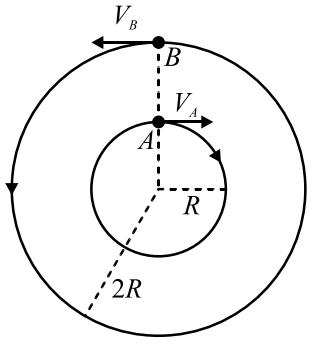
(1) 2 (2) 3
(3) 4 (4) 6

41. Choose the option containing **correct** statement/s.

A. In the stress-strain curve for steel, the Hooke's law is valid till proportionality limit.
B. Bulk modulus is defined for all three states of matter i.e. solid, liquid and gas.
C. A uniform rod having mass is suspended vertically on the ceiling. The normal stress in the rod at every cross-section changes as we move along the rod.

(1) A, B and C
(2) B and C only
(3) A and C only
(4) B only

42. Two particles A and B are moving in opposite sense on concentric circles with equal angular speed ω as shown below. What is the magnitude of relative linear velocity of B w.r.t A at the instant shown?



(1) ωR (2) $3\omega R$
(3) $2\omega R$ (4) zero

43. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Viscosity of liquids decreases with rise in temperature.

Reason R: Cohesive forces in liquid increases with a rise in temperature.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false
(2) A is false but R is true
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

44. In the table below, **List-I** contains the names of Kepler's Laws and **List-II** contains their corresponding Mathematical Expressions or Physical Principles. Match the items from **List-I** with the **correct** descriptions in **List-II**. (Symbols have usual meanings)

List-I (Laws)		**List-II (Formulas & Principles)**	
(A)	Kepler's First Law	(I)	$\frac{dA}{dt} = \frac{L}{2m}$ = constant
(B)	Kepler's Second Law	(II)	$T^2 \propto a^3$
(C)	Kepler's Third Law	(III)	Law of Orbits (Elliptical path)

Choose the **correct** answer from the options given below:

(1) A-III, B-I, C-II
(2) A-II, B-I, C-III
(3) A-I, B-III, C-II
(4) A-II, B-III, C-I

45. A balloon initially at rest starts rising from the ground with an upward acceleration 4 m/s^2. After 10 seconds, a stone is dropped from the balloon. The speed of stone for a person standing on ground 3 seconds after the stone is dropped is:

(Take g = 10 m/s^2)

(1) 20 m/s
(2) 10 m/s
(3) 15 m/s
(4) 40 m/s

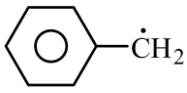
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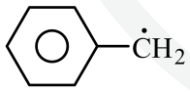
46. 11.2 L of which gases will weigh 14 g at STP?
I. CO_2
II. N_2
III. NO_2
IV. CO
(1) I and IV
(2) II and III
(3) II and IV
(4) I and III

47. Which of the following electron transitions in H-atom will require minimum energy?
(1) n = 1 to n = 2
(2) n = 2 to n = 3
(3) n = 3 to n = 4
(4) n = 4 to n = 5

48. Element with Z = 51 belongs to which group number and period number in modern periodic table?
(1) group number = 12, period number = 6
(2) group number = 13, period number = 7
(3) group number = 15, period number = 5
(4) group number = 14, period number = 5

49. Given below are two statements

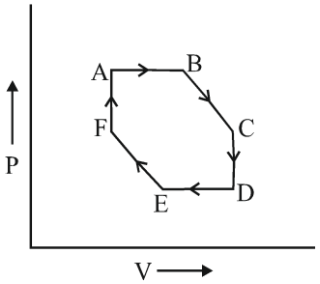
Statement 1: $CH_2=CH-\dot{C}H_2$ is more stable than  .

Statement 2 :  is more stable than $CH_2=CH-\dot{C}H_2$.

In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

50. Due to inert pair effect, 13th group element shows which oxidation state?
(1) +1
(2) +2
(3) +3
(4) +5

51. For the given graph, which among the following statement is **correct**?

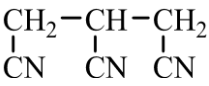


(I) A → B represents isobaric process.
(II) C → D represents isochoric process.
(III) D → E represents isothermal process.
(1) II and III
(2) I and III
(3) I, II and III
(4) I and II

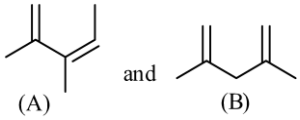
52. Number of lone pair present on central atom in I_3^+ and I_3^- respectively are x and y, calculate y-x:
(1) 3 (2) 2
(3) 5 (4) 1

53. Which among the following compounds can act as oxidising as well as reducing agent?
(1) CO_2 (2) H_3PO_4
(3) HCl (4) SO_2

54. In which of the following equilibria, $K_p = K_c(RT)^{-2}$?
(1) $N_2O_4(g) \rightleftharpoons 2NO_2(g)$
(2) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$
(3) $SO_2Cl_2(g) \rightleftharpoons SO_2(g) + Cl_2(g)$
(4) $CO_2(g) + C(s) \rightleftharpoons 2CO(g)$

55. The IUPAC name of  is:
(1) Propane-1,2,3-tricarbonitrile
(2) 3-Cyanopentane-1,5-dinitrile
(3) 3,4-dicyanobutanenitrile
(4) 1,2-dicyanobutanenitrile

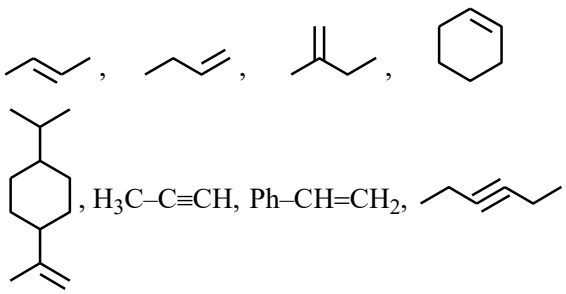
56. The relation between (A) and (B) is;



(1) Chain Isomers
(2) Homologues
(3) Identical
(4) Position Isomers

[Contd...

57. Examine the structural formulas of following compounds



Find how many compounds will not produce CO_2 on oxidative ozonolysis.

(1) 3
(2) 6
(3) 5
(4) 7

58. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Law of conservation of mass is valid only for radioactive reactions.

Reason R: According to law of conservation of mass, matter can neither be created nor be destroyed.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

59. Which of the following statements form a part of Bohr's model of hydrogen atom?

(1) Energy of the electrons in the orbits are quantized.
(2) Electrons are fixed around the nucleus.
(3) The electron velocity and position in the orbit cannot be determined simultaneously.
(4) the emission of electrons from a metal surface when exposed to light or electromagnetic radiation of a sufficiently high frequency.

60. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:

Assertion A: The chemistry of early lanthanoids is more complicated than the corresponding actinoids.

Reason R: It is due to large number of oxidation states possible for these actinoids elements.

In the light of above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

61. Match **List I** with **List II.**

List I (Process)		List II (Heat change (q))	
(A)	Isothermal	(I)	–w
(B)	Isobaric	(II)	0
(C)	Adiabatic	(III)	$nC_n\Delta T$
(D)	Isochoric	(IV)	ΔU

Choose the **correct** answer from the options given below:

(1) A-II, B-I, C-IV, D-III
(2) A-I, B-III, C-II, D-IV
(3) A-II, B-III, C-I, D-IV
(4) A-I, B-II, C-IV, D-III

62. Which of the following bond formation by the overlapping of orbitals is **incorrect** (assuming x-axis is internuclear axis)?

(1) $s + p_z \rightarrow \sigma$-Bond formation
(2) $p_y + p_y \rightarrow \pi$-Bond formation
(3) $p_x + p_y \rightarrow$ Zero overlap
(4) $p_z + p_z \rightarrow \pi$-Bond formation

63. Consider the reaction:

$$xFe^{2+} + yCr_2O_7^{2-} + zH^+ \rightarrow xFe^{3+} + 2yCr^{3+} + \frac{z}{2}H_2O$$

Calculate z + x – y for the balanced reaction is

(1) 19 (2) 22
(3) 20 (4) 21

64. Given below are two statements:

Statement I: If K_c is very small, the reversible reaction proceeds nearly to completion.

Statement II: If K_c is very large, the reversible reaction proceeds rarely.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

[Contd...

65. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Cyanides (R–C≡N) and isocyanides (R–N≡C) of the same alkyl group are position isomers.

Reason R: In these compounds the connectivity of carbon and nitrogen with the alkyl group is different even though the overall formula and atoms present are the same.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

66. Match the List-I with List-II.

List-I		List-II	
(A)	Molarity of pure water	(I)	1 M
(B)	Molarity of 1 mole of solute in 1 L of solution	(II)	55.55 M
(C)	Molarity of 49 g of H_2SO_4 dissolved in 100 mL of solution	(III)	5 M
(D)	Molarity of 1 mole of NaOH dissolved in 100 g of solution (ρ of solution = 1.2 g/mL)	(IV)	12 M

Choose the **correct** answer from the options given below;

(1) A-II, B-III, C-I, D-IV
(2) A-III, B-I, C-II, D-IV
(3) A-IV, B-III, C-I, D-II
(4) A-II, B-I, C-III, D-IV

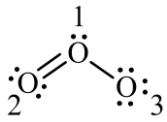
67. The ratio of kinetic energy to the potential energy of electron in ground state of H-atom is:

(1) –1
(2) –2
(3) $\frac{1}{2}$
(4) $-\frac{1}{2}$

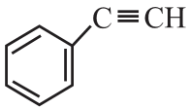
68. First trans-uranic element is:

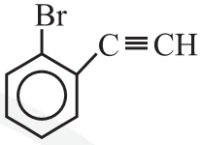
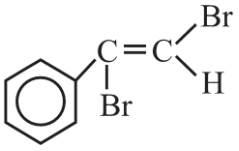
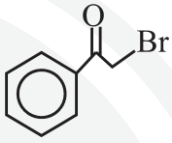
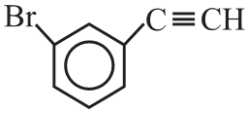
(1) Np
(2) No
(3) Es
(4) Lr

69. The formal charges on oxygen which are labelled as 1,2 and 3 respectively are:



(1) +1, 0, –1
(2) +1 , –1, 0
(3) 0, +1, –1
(4) –1, 0, +1

70.  $\xrightarrow{Br_2, H_2O}$ Major product [A] is:

(1) 
(2) 
(3) 
(4) 

71. In which of the following, there are five significant figures?

I. 0.00250
II. 0.63000
III. 7.0004
IV. 1.0000

(1) I and III
(2) III only
(3) II, III and IV
(4) III and IV

72. Unit of wave number and wavelength respectively are:

(1) m and m^{-1}
(2) $m^2$and m
(3) m^{-1} and m
(4) Unitless and m

73. Which of the following statement is **correct**?

I. Aromatic compounds containing a benzene ring are called benzenoids.
II. Aromatic compounds not containing a benzene ring (but aromatic) are called nonbenzenoids.
III. Naphthalene is non-benzenoid because it has two rings.
IV. Benzene and toluene are benzenoids.

(1) I, II and IV only
(2) I and III only
(3) II and III only
(4) I, II and III only

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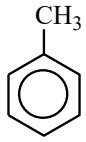
74. Match **List I** with **List II.**

List I (Element)		List II (Metallic radii)	
(A)	Boron (B)	(I)	167 pm
(B)	Aluminum (Al)	(II)	88 pm
(C)	Gallium (Ga)	(III)	143 pm
(D)	Indium (In)	(IV)	135 pm

Choose the **correct** answer from the options given below:

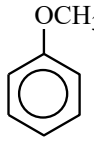
(1) A-II, B-I, C-IV, D-III
(2) A-I, B-III, C-II, D-IV
(3) A-II, B-III, C-IV, D-I
(4) A-I, B-II, C-IV, D-III

75. Decreasing order of reactivity towards electrophile substitution for the following compounds is:

A. 

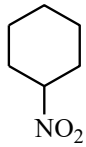
B. 

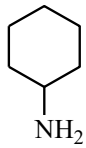
C. 

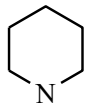
D. 

(1) C > B > A > D
(2) D > A > B > C
(3) A > D > B > C
(4) D > A > C > B

76. Nitrogen can be estimated by Kjeldahl's method for which of the following compound?

(1) 

(2) 

(3) 

(4) 

77. Consider the statements:

I. Germanium forms stable compounds in +4 state and only a few in +2 state.
II. Tin forms compounds in both +2 and +4 states, and Sn(II) acts as a reducing agent.
III. Lead compounds in +2 state are more stable, and Pb(IV) compounds act as strong oxidising agents.
IV. The oxidising character of Pb(IV) is because Pb(IV) is more stable than Pb(II).

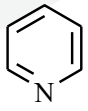
(1) I, II and IV only
(2) I, II and III only
(3) II and III only
(4) I and III only

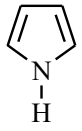
78. Consider the statements:

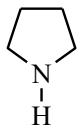
I. Variables like p, V, T are called state variables/state functions because their values depend only on the state, not on the path.
II. If a system returns to the same initial state, all state variables regain their original values.
III. Whether a variable is a state function can be decided by checking if its change depends on the process followed.
IV. The value of a state variable at a given instant depends on how the system reached that state.

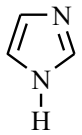
(1) I, II and III only
(2) I and IV only
(3) II and IV only
(4) III and IV only

79. The increasing order of basicity of the following compounds:

A. 

B. 

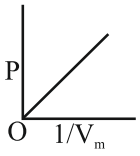
C. 

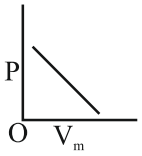
D. 

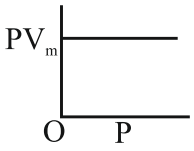
(1) A < B < C < D
(2) D < A < B < C
(3) B < A < C < D
(4) B < A < D > C

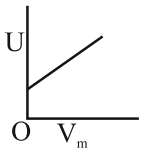
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80. The combination of plots which do not represent isothermal expansion of an ideal gas is:

A. 

B. 

C. 

D. 

(1) B and D
(2) A and C
(3) B and C
(4) A and D

81. In **paper chromatography**, identify the statement that is **incorrect** regarding the stationary phase.
(1) Water present in the pores of the paper forms the stationary phase.
(2) Paper sheet itself alone is the stationary phase.
(3) Stationary phase in paper chromatography is water held in the paper pores.
(4) The mobile phase is the solvent that moves through the paper.

82. In acidified aqueous solution, aluminium chloride forms an ion whose geometry is:
(1) trigonal bipyramidal
(2) octahedral
(3) tetrahedral
(4) Square planar

83. The oxidation state of chromium in chromate and dichromate salts is respectively:
(1) +6 and +3
(2) +6 and +6
(3) +3 and +6
(4) +5 and +6

84. In which medium will the solubility of AgCl be maximum?
(1) Deionised water
(2) 0.01 M KCl
(3) 0.01 M HCl
(4) 0.01 M $AgNO_3$

85. In SO_2, NO_2^- and N_3^-, the electron-pair geometry around the central atom is respectively:
(1) trigonal planar, linear, trigonal planar
(2) trigonal planar, trigonal planar, linear
(3) linear, trigonal planar, linear
(4) bent, linear, bent

86. The extra stability of a half-filled subshell is due to:
(A) Symmetrical distribution of electrons
(B) Smaller coulombic repulsion energy
(C) The presence of electrons with the same spin in non-degenerate orbitals
(D) Larger exchange energy
(E) Relatively smaller shielding of electrons by one another

Identify the set that lists the statements which are **not** considered valid reasons for this extra stability.
(1) C only
(2) E only
(3) C and E only
(4) B only

87. Given K_a for $CH_3COOH = 1.8 \times 10^{-5}$ and K_b for $NH_4OH = 1.8 \times 10^{-5}$, ammonium acetate solution is:
(1) acidic (pH < 7)
(2) basic (pH > 7)
(3) neutral (pH = 7)
(4) cannot be predicted from K_a and K_b

88. Among the species $N_2, N_2^+, N_2^-, N_2^{2-}, O_2, O_2^+, O_2^-, O_2^{2-}$, the number of paramagnetic species is:
(1) 2
(2) 4
(3) 6
(4) 8

89. Which mixing of 1 M base and 1 M acid releases the maximum heat (in kJ) (assuming same initial temperature and negligible heat loss)?
(1) 45 mL CH_3COOH and 25 mL NaOH
(2) 50 mL HCl and 20 mL NaOH
(3) 30 mL CH_3COOH and 30 mL NaOH
(4) 30 mL HCl and 30 mL NaOH

90. Which among the following electronic configurations do **not** belong to main group elements?
A. $[Ne]3s^1$
B. $[Ar]3d^3 4s^2$
C. $[Kr]4d^{10} 5s^2 5p^5$
D. $[Ar]3d^{10} 4s^1$

Choose the **correct** answer from the option given below:
(1) B and D only
(2) A and C only
(3) D and C only
(4) A, C and D only

[Contd...

91. The understanding that all present-day organisms are related to each other and to all organisms that ever lived, resulted in:
(1) an anthropocentric view of biology.
(2) limited progress in biological knowledge.
(3) cultural movements for conservation of biodiversity.
(4) the development of the binomial nomenclature system.

92. In Engelmann's experiment, which of the following combinations were used?
(1) Anaerobic bacteria and a variegated leaf
(2) Aerobic bacteria and *Cladophora*
(3) A variegated leaf treated with KOH and bell jar
(4) An aquatic plant in the dark with a supply of H_2S

93. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): Meiosis is called reduction division.
Reason (R): Meiosis reduces the chromosome number by half in the production of haploid daughter cells.
In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

94. In a dorsiventral leaf, the tissue between the upper and the lower epidermis is;
(1) made of barrel-shaped cells.
(2) made of dead cells.
(3) made of only one type of cells.
(4) primarily associated with photosynthesis.

95. Match **List-I** with **List-II**.

List-I (Disease)		**List-II (Type of pathogen)**	
(A)	Rust and smut in plants	(I)	Protist
(B)	Malaria in humans	(II)	Bacterium
(C)	Canker in citrus	(III)	Fungus
(D)	Mosaic in tobacco	(IV)	Virus

Choose the *most appropriate* answer from the options given below.
(1) A-III, B-I, C-II, D-IV
(2) A-IV, B-II, C-I, D-III
(3) A-III, B-IV, C-II, D-I
(4) A-II, B-I, C-IV, D-III

96. Identify the **correct** statement about the simplest plant growth hormone.
(1) More than 100 types of this PGR have been isolated from plants and fungi.
(2) It is not produced naturally by plants.
(3) Ethephon is the most widely used compound as its source.
(4) It was first isolated from human urine.

97. Given below are two statements:
Statement I: The plasma membrane is selectively permeable, allowing some molecules to pass without using energy, a process termed passive transport.
Statement II: Osmosis is a form of active transport.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Both Statement I and Statement II are incorrect.
(4) Statement I is incorrect but Statement II is correct.

98. Decapitation in plants is done to promote the growth of;
(1) apical buds.
(2) roots.
(3) secondary vascular tissues.
(4) axillary buds.

99. Identify the **correctly** matched pair regarding characteristics of members of the mentioned algal class.
(1) Rhodophyceae - 2-8 equal, apical flagella
(2) Chlorophyceae - Possess r-phycoerythrin
(3) Phaeophyceae - Laminarin or mannitol as stored food
(4) Chlorophyceae - Floridean starch as stored food

100. The RQ for a fatty acid would be;
(1) <1
(2) >1
(3) 0
(4) exactly 1

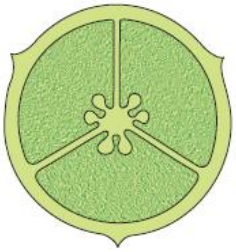
101. Given below are two statements:
Statement I: In monocotyledonous roots, the pith is large and well developed.
Statement II: Monocotyledonous roots undergo significant secondary growth.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

102. Arrange the following taxonomic categories of mango in ascending order of hierarchy.
A. Plantae
B. Sapindales
C. Anacardiaceae
D. *Mangifera*
E. Angiospermae
Choose the **correct** option.
(1) E → D → C → A → B
(2) D → C → B → E → A
(3) A → B → C → D → E
(4) B → A → C → D → E

103. During fixation of atmospheric CO_2 in the Calvin cycle, RuBisCO primarily acts as a/an;
(1) oxygenase. (2) hydrolase.
(3) isomerase. (4) carboxylase.

104. Identify the **correct** statements with respect to the pachytene stage of meiosis.
A. It is the stage where the four chromatids of each bivalent chromosome become distinct.
B. Crossing over occurs between sister chromatids of homologous chromosomes.
C. It is characterized by the appearance of recombination nodules.
D. Recombination between homologous chromosomes is completed by the end of this stage.
E. This stage is relatively short-lived compared to leptotene and zygotene.
Choose the **correct** option.
(1) A, C and D
(2) A, B and E
(3) B, D and E
(4) A, C and E

105. The given diagram represents a type of placentation. Identify the type and an example where it is found.



(1) Marginal, Pea
(2) Axile, China rose
(3) Parietal, Mustard
(4) Free central, *Dianthus*

106. Methanogens are classified as;
(1) Eubacteria.
(2) Cyanobacteria.
(3) Archaebacteria.
(4) Mycoplasma.

107. Identify the plants that are considered as amphibians of plant kingdom.
A. *Polytrichum*
B. *Marchantia*
C. *Adiantum*
D. *Cedrus*
E. *Sphagnum*
Choose the **correct** option.
(1) A and D
(2) B and C
(3) C, D and E
(4) A, B and E

108. Match **List-I** with **List-II**.

List-I (Organelle)		**List-II (Characteristic Feature)**	
(A)	Golgi apparatus	(I)	Bounded by tonoplast
(B)	Lysosome	(II)	Cisternae stacked parallel to each other
(C)	Vacuole	(III)	Rich in hydrolytic enzymes
(D)	Ribosome	(IV)	Non-membrane bound organelle

Choose the *most appropriate* answer from the options given below.
(1) A-II, B-III, C-I, D-IV
(2) A-II, B-I, C-IV, D-III
(3) A-IV, B-III, C-I, D-II
(4) A-I, B-II, C-III, D-IV

[Contd...

109. The final acceptor of electrons in the mitochondrial ETS is;
(1) Cytochrome *c*.
(2) NAD^+.
(3) Water.
(4) Oxygen.

110. Identify the **correct** statement about arithmetic growth.
(1) It produces a sigmoid when growth is plotted against time.
(2) Observed in most systems with limited nutrient supply.
(3) Following mitotic cell division, only one daughter cell continues to divide.
(4) The initial growth is slow, followed by a rapid exponential phase and then a stationary phase.

111. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): In dicot plants, the lateral roots develop from the cells of pericycle.
Reason (R): The vascular cambium forms from the cells of pericycle in dicot root.
In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

112. Given below are two statements:
Statement I: In C_4 plants, the primary CO_2 fixation product is a 4-carbon organic acid.
Statement II: C_4 plants use the Calvin cycle as the main biosynthetic pathway.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

113. In the 24-hour average duration of a human cell cycle, the resting phase lasts for:
(1) about an hour.
(2) exactly 12 hours.
(3) more than 95% of the duration.
(4) less than 50% of the duration.

114. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): The two-kingdom classification system was found to be inadequate.
Reason (R): The two-kingdom classification system divided kingdom Monera into two domains.
In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is false but (R) is true.
(4) (A) is true but (R) is false.

115. Identify the **incorrectly** matched pair.
(1) Species - Lowest taxonomical category of all organisms
(2) Division - Rank equivalent to phylum
(3) Order - Included related families
(4) Genus - Only one organism assigned for each genus

116. In *Rhizophora*, the roots modified for respiration are called:
(1) Prop roots.
(2) Stilt roots.
(3) Pneumatophores.
(4) Fibrous roots.

117. Given below are two statements:
Statement I: In gymnosperms, the male and female gametophytes have an independent, free-living existence.
Statement II: The female gametophyte in gymnosperms is multicellular.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is incorrect but Statement II is correct.
(3) Statement I is correct but Statement II is incorrect.
(4) Both Statement I and Statement II are incorrect.

118. How many of the following statements regarding the nucleus are correct?

A. The perinuclear space is a barrier between materials in the nucleus and the nucleoplasm.
B. The outer nuclear membrane is continuous with the rough endoplasmic reticulum.
C. Nuclear pores are passages for the movement of RNA and protein molecules in one direction only, from nucleus to cytoplasm.
D. A single human cell has approximately two metres of DNA distributed among its forty-six chromosomes.
E. The nucleolus is a membrane-bound structure active in ribosomal RNA synthesis.

Choose the **correct** option.

(1) Two only
(2) One only
(3) Three only
(4) Four only

119. If yeast cells are grown in anaerobic conditions with glucose as the sole respiratory substrate, which of the following enzymes is likely not to be involved in respiration?

(1) Hexokinase
(2) Alcohol dehydrogenase
(3) Pyruvic acid decarboxylase
(4) Citrate synthase

120. The fundamental tissue system in plants;

(1) is comprised of the complex tissues.
(2) is also called the ground tissue system.
(3) is constituted by all tissues except epidermis.
(4) bears trichomes and root hairs.

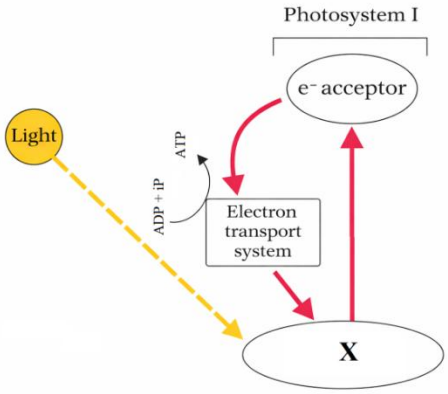
121. The plant growth hormone that is a derivative of carotenoids is;

(1) Auxin.
(2) Gibberellin.
(3) Ethylene.
(4) Abscisic acid.

122. _____ reported that the curly top virus spreads through the phloem.

(1) Katherine Esau
(2) Melvin Calvin
(3) J. A. Bassham
(4) Ernst Mayr

123. Choose the **correct** statement about the type of photophosphorylation shown below.



(1) The possible location of this process is granal lamellae.
(2) This type of photophosphorylation occurs when light of wavelength less than 650 nm is available.
(3) The reaction centre of 'X' is a molecule of chlorophyll *a*.
(4) The electrons lost from PS I are compensated by water.

124. Identify the **correct** statements regarding fungi.

A. The cell walls of fungi are composed of chitin and polysaccharides.
B. All fungi are filamentous, consisting of long, slender thread-like structures called hyphae.
C. Fungi prefer to grow in cold and dry places.
D. The fusion of two nuclei in the fungal sexual cycle is called plasmogamy.
E. In ascomycetes and basidiomycetes, a dikaryotic stage (n+n) occurs during sexual reproduction.

Choose the **correct** option.

(1) A, B, and E (2) C and D only
(3) B, C, and D (4) A and E only

125. Match **List-I** with **List-II**.

List-I (Condition of stamen)		**List-II (Example)**	
(A)	Epiphyllous	(I)	China rose
(B)	Epipetalous	(II)	Lily
(C)	Monoadelphous	(III)	*Citrus*
(D)	Polyadelphous	(IV)	Brinjal

Choose the *most appropriate* answer from the options given below.

(1) A-II, B-IV, C-I, D-III
(2) A-II, B-I, C-IV, D-III
(3) A-III, B-IV, C-II, D-I
(4) A-I, B-II, C-III, D-IV

[Contd...

126. Arrange the following events of mitosis in the correct sequence.

A. Chromosomes are moved to the spindle equator.

B. Centromere splits and chromatids separate.

C. Chromosomal material condenses.

D. Chromosomes cluster at opposite poles.

Choose the **correct** option.

(1) A → C → D → B

(2) C → A → B → D

(3) C → D → A → B

(4) D → B → A → C

127. Which of the following is **not** present in grasses?

(1) Dumb-bell shaped guard cells

(2) Intercalary meristems

(3) Spongy parenchyma

(4) Bulliform cells

128. In which of the following plants, an event precursor to seed habit is observed?

(1) *Pinus*

(2) *Salvinia*

(3) *Salvia*

(4) *Wolffia*

129. The cell organelle that helps in glycogen synthesis is;

(1) Mitochondrion.

(2) Chloroplast.

(3) Peroxisome.

(4) Endoplasmic reticulum.

130. Match **List-I** with **List-II**.

List-I (Process)		**List-II (Product)**	
(A)	Glycolysis	(I)	Lactic acid
(B)	Calvin cycle	(II)	Phosphoglycolate and phosphoglycerate
(C)	Photorespiration	(III)	Sugar
(D)	Fermentation in animal muscle cells	(IV)	Pyruvic acid

Choose the *most appropriate* answer from the options given below.

(1) A-IV, B-III, C-II, D-I

(2) A-IV, B-II, C-I, D-III

(3) A-II, B-III, C-IV, D-I

(4) A-I, B-IV, C-II, D-III

131. Phototropism was observed in coleoptiles of canary grass by;

(1) F. W. Went.

(2) E. Kurosawa.

(3) Charles Darwin and Francis Darwin.

(4) F. Skoog and his co-workers.

132. Viroids differ from viruses in:

(1) having a protein coat.

(2) being larger in size.

(3) having DNA as the genetic material.

(4) having free RNA without a protein coat.

133. Read the following characteristics w.r.t the Poaceae family.

A. Fruit is caryopsis.

B. Also called cereal family.

C. The members are dicots.

D. Flower is trimerous.

E. Fruit has more than one seeds.

The **incorrect** statements are;

(1) A and D

(2) C and E

(3) B and C

(4) A and E

134. In a chromatogram, yellow colour is shown by;

(1) chlorophyll *a*.

(2) chlorophyll *b*.

(3) carotenoid.

(4) xanthophyll.

135. In algae, zoospores are:

(1) motile.

(2) always pyriform shaped.

(3) meant for sexual reproduction.

(4) structures for vegetative reproduction.

[Contd...

136. The gap between the pre-synaptic and post-synaptic membranes is known as the:
(1) Synaptic knob
(2) Synaptic vesicle
(3) Synaptic cleft
(4) Receptor site

137. Which of the following amino acids is a basic amino acid ?
(1) Glutamic acid
(2) Serine
(3) Valine
(4) Lysine

138. Match **List-I** with **List-II**.

List-I (Animal)		List-II (Characteristic feature)	
(A)	*Pila*	(I)	Comb plates
(B)	*Bombyx*	(II)	Water vascular system
(C)	*Pleurobrachia*	(III)	Radula
(D)	*Asterias*	(IV)	Malpighian tubules

Choose the **correct** answer from the options given below:
(1) A-II, B-III, C-I, D-IV
(2) A-III, B-II, C-I, D-IV
(3) A-III, B-IV, C-I, D-II
(4) A-III, B-II, C-IV, D-I

139. Identify the **correctly** matched pair regarding the cockroach anatomy.
(1) Tegmina - Transparent, membranous hind wings used in flight
(2) Hepatic caeca - Junction of midgut and hindgut
(3) Anal styles - Present in both male and female cockroaches
(4) Mushroom gland - Accessory reproductive gland in 6th-7th abdominal segments of males

140. A student finds a marine invertebrate with a soft, unsegmented body protected by a calcareous shell. It has a sensory tentacle and a file-like rasping organ for feeding. To which class/phylum does this animal most likely belong?
(1) Echinodermata
(2) Mollusca
(3) Arthropoda
(4) Hemichordata

141. Consider the following statements regarding the formed elements of blood:
A. Erythrocytes are the most abundant cells in blood, with a count of 5 to 5.5 billion per mm^3.
B. Neutrophils and lymphocytes are phagocytic cells that destroy foreign organisms.
C. Basophils secrete histamine, serotonin, and heparin and are involved in inflammatory reactions.
D. Eosinophils resist infections and are associated with allergic reactions.
E. Platelets are cell fragments produced from thrombocytes.
Which of the above statements are **incorrect**?
(1) C and D only
(2) A, B and E only
(3) B, C, and D only
(4) A, C, and E only

142. Given below are two statements:
Statement I: Atrial Natriuretic Factor (ANF) acts as a check on the renin-angiotensin mechanism by causing vasoconstriction.
Statement II: The mechanism of concentration of filtrate is primarily dependent on the counter-current pattern of the loop of Henle and *vasa recta*.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

143. The transition stage of a substrate during an enzymatic reaction is:
(1) Stable and of lower energy than the product.
(2) Unstable and transient with high internal energy.
(3) Permanent and requires no energy input.
(4) Always identical to the structure of the end product.

144. A chronic smoker visits a clinic complaining of shortness of breath. Clinical tests reveal that his alveolar walls are damaged, leading to a decreased respiratory surface. He is **most** likely suffering from:
(1) Asthma (2) Emphysema
(3) Silicosis (4) Fibrosis

[Contd...

145. Identify the **incorrect** match regarding the hormones of the Gastro-Intestinal Tract:
(1) Gastrin - Stimulates secretion of HCl and pepsinogen.
(2) Secretin - Stimulates secretion of water and bicarbonate ions from the pancreas.
(3) Cholecystokinin (CCK) - Acts on both pancreas and gall bladder to stimulate secretion of enzymes and bile juice.
(4) Gastric Inhibitory Peptide (GIP) - Stimulates gastric secretion and motility.

146. Which of the following statements regarding animal tissues is **incorrect**?
(1) Smooth muscle fibres are fusiform and involuntary in nature.
(2) Neuroglia make up more than one-half the volume of neural tissue in our body.
(3) In adult vertebrates, all cartilages are unexceptionally replaced by bones.
(4) Dense regular connective tissue contains collagen fibres in rows between parallel bundles of fibres.

147. Choose the **correct** statement regarding the functions of the renal tubules:
(1) Reabsorption is minimum in the Proximal Convoluted Tubule (PCT) to maintain high filtrate volume.
(2) The descending limb of Henle's loop is permeable to electrolytes but impermeable to water.
(3) The ascending limb of Henle's loop allows transport of electrolytes actively or passively and is impermeable to water.
(4) Conditional reabsorption of Na^+ and water takes place primarily in the loop of Henle.

148. Match **List-I** with **List-II.**

List-I		List-II	
(A)	Tight junctions	(I)	Involved in forming a diffusion boundary
(B)	Adhering junctions	(II)	Facilitate rapid transfer of ions and small molecules
(C)	Gap junctions	(III)	Perform cementing to keep neighboring cells together
(D)	Squamous epithelium	(IV)	Help to stop substances from leaking across a tissue

Choose the **correct** answer from the options given below:
(1) A-III, B-IV, C-I D-II
(2) A-IV, B-III, C-I, D-II
(3) A-III, B-IV, C-II, D-I
(4) A-IV, B-III, C-II, D-I

149. Which of the following **correctly** describes the volume changes involved during inspiration?
(1) Contraction of diaphragm increases thoracic volume in the dorso-ventral axis
(2) Contraction of external inter-costal muscles increases thoracic volume in the antero-posterior axis
(3) Contraction of diaphragm increases thoracic volume in the antero-posterior axis
(4) Relaxation of abdominal muscles increases pulmonary volume

150. Which of the following is a **correctly** matched pair?
(1) *Ophiura* – Sea urchin
(2) *Antedon* – Sea lily
(3) *Echinus* – Brittle star
(4) *Cucumaria* – Star fish

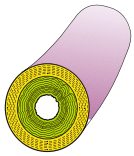
151. Which of the following statements is **correct**?
(1) Inorganic catalysts work efficiently at high temperatures while enzymes are denatured.
(2) Competitive inhibitors resemble the enzyme in its molecular structure
(3) Co-enzymes are permanently bound to the apoenzyme to make it active.
(4) All enzymes are proteinaceous in nature.

152. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Cellulose cannot hold iodine molecules and remains colorless in an iodine test.
Reason R: Cellulose does not contain complex helices.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

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153. Choose the **correct** statement regarding Phylum Ctenophora:
(1) They exhibit both asexual and sexual modes of reproduction.
(2) Digestion is entirely extracellular.
(3) They are exclusively marine, diploblastic, and exhibit bioluminescence.
(4) Fertilisation is internal with direct development.

154. In the given diagram of a blood vessel, **correctly** identify the type of muscle present in its wall?



(1) Smooth muscle (2) Skeletal muscle
(3) Cardiac muscle (4) Voluntary muscle

155. A clinical patient presents with rapid spasms (wild contractions) in their muscles. This condition is **most** likely linked to:
(1) Inflammation of the joints due to uric acid accumulation.
(2) Decreased levels of estrogen leading to reduced bone mass.
(3) Low Ca^{2+} levels in the body fluids causing Tetany.
(4) A genetic disorder causing progressive degeneration of skeletal muscle.

156. Regarding the micturition reflex, select the **correct** sequence of events:
(1) CNS sends motor signals → Relaxation of bladder muscles → Contraction of urethral sphincter.
(2) Stretch receptors on bladder wall send signals to CNS → Motor signals trigger contraction of smooth muscles of bladder → Relaxation of urethral sphincter.
(3) Increase in urine volume → Inhibition of osmoreceptors → Voluntary contraction of bladder muscles.
(4) CNS sends sensory signals → Contraction of ureters → Opening of the urinary bladder.

157. Which of the following centers are located in the medulla oblongata?
(1) Respiratory center
(2) Cardiovascular reflexes center
(3) Gastric secretion center
(4) All of these

158. Match **List-I** with **List-II** w.r.t. mouthparts in cockroach.

List-I		List-II	
(A)	Labrum	(I)	Lower lip
(B)	Labium	(II)	Median flexible lobe acting as a tongue
(C)	Hypopharynx	(III)	Upper lip
(D)	Mandibles	(IV)	Possess grinding and incising regions

Choose the **correct** answer from the options given below:
(1) A-I, B-III, C-IV, D-II
(2) A-III, B-I, C-IV, D-II
(3) A-III, B-I, C-II, D-IV
(4) A-III, B-IV, C-II, D-I

159. Which of the following **correctly** describes the structural changes in a muscle fiber during a state of maximal contraction?
(1) The 'Z' lines move further apart, increasing the sarcomere length.
(2) The 'H' zone disappears as thin filaments from opposite sides move towards the center.
(3) The 'A' band expands to cover the entire length of the myofibril.
(4) The concentration of Ca^{2+} in the sarcoplasmic reticulum reaches its peak.

160. Match **List-I** with **List-II.**

List-I		List-II	
(A)	Uremia	(I)	Inflammation of glomeruli of kidney
(B)	Renal calculi	(II)	Accumulation of urea in blood due to kidney malfunction
(C)	Glomerulonephritis	(III)	Process of removing urea from blood using an artificial kidney
(D)	Hemodialysis	(IV)	Mass of crystallized salts (oxalates, etc.) formed within the kidney

Choose the **correct** answer from the options given below:
(1) A-II, B-IV, C-I, D-III
(2) A-II, B-III, C-I, D-IV
(3) A-III, B-I, C-II, D-IV
(4) A-III, B-I, C-IV, D-II

[Contd...

161. Identify the **correctly** matched pair regarding tissue location and type.
(1) Proximal convoluted tubule (PCT) — Ciliated columnar epithelium
(2) Walls of blood vessels — Ciliated epithelium
(3) Alveoli — Simple cuboidal epithelium
(4) Lining of stomach — Columnar epithelium

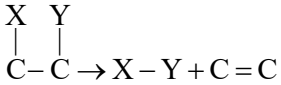
162. A student analyzes a cell extract and finds a biomolecule that is found in the acid-insoluble fraction but has a molecular weight of approximately 750 Da. Upon further testing, it is found to contain a phosphate group and is a component of the cell membrane. This molecule is **most** likely:
(1) A typical protein
(2) A complex polysaccharide
(3) Lecithin
(4) Adenylic acid

163. Match **List-I** with **List-II.**

List-I		List-II	
(A)	Fibrinogen	(I)	Defense mechanisms of the body
(B)	Globulins	(II)	Blood clotting or coagulation
(C)	Albumins	(III)	Plasma without clotting factors
(D)	Serum	(IV)	Osmotic balance

Choose the **correct** answer from the options given below:
(1) A-III, B-IV, C-II, D-I
(2) A-I, B-II, C-IV, D-III
(3) A-II, B-III, C-I, D-IV
(4) A-II, B-I, C-IV, D-III

164. The type of reaction shown by given image is catalysed by which of the following enzymes?



(1) Lyases
(2) Transferases
(3) Hydrolases
(4) Ligases

165. Which of the following is known as the "Fighting fish"?
(1) *Exocoetus*
(2) *Clarias*
(3) *Pterophyllum*
(4) *Betta*

166. Which of the following gland is responsible for secretion of growth hormone?
(1) Neurohypophysis
(2) Adenohypophysis
(3) Thymus
(4) Parathyroid gland

167. Which of the following statements regarding Phylum Chordata is **correct**?
(1) In Urochordata, the notochord extends from head to tail and is persistent throughout life.
(2) In Cephalochordata, the notochord is present only in the larval tail.
(3) All vertebrates are chordates, but all chordates are not vertebrates.
(4) Central nervous system is ventral, solid, and double in chordates.

168. Select the **incorrect** statement regarding the frog's anatomy.
(1) The frog never drinks water but absorbs it through the skin.
(2) Male frogs possess copulatory pads on first digits of their hindlimbs.
(3) The heart is three-chambered with two atria and one ventricle.
(4) The frog is a ureotelic animal, excreting urea as its primary nitrogenous waste.

169. Regarding a glycogen molecule, which of the following is **true**?
(1) The right end is the reducing end.
(2) It is a heteropolymer of fructose.
(3) Both ends are non-reducing.
(4) The left end is the reducing end.

170. The total number of bones in both forelimbs in humans is:
(1) 80 (2) 30
(3) 40 (4) 60

171. Given below are two statements:
Statement I: Frogs undergo aestivation and hibernation during peak winters and summers respectively.
Statement II: Frogs are poikilotherms, meaning they can maintain a constant body temperature regardless of environmental changes.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

[Contd...

172. Select the **correct** statements regarding the human skeletal system and its articulations:

A. The human skull is dicondylic as it articulates with the axis via two occipital condyles.
B. The 8th, 9th, and 10th pairs of ribs are called vertebrochondral ribs.
C. Each half of the pectoral girdle consists of a clavicle and a scapula.
D. The acromion is a flat, expanded process of the scapula that articulates with the humerus.
E. The pelvic girdle is formed by two coxal bones, each resulting from the fusion of the ilium, ischium, and pubis.

(1) A, B, and D only (2) A, B, C, and E only
(3) B, C and E only (4) A, B, C, D and E

173. Which of the following **correctly** describes heart failure?

(1) It is the same as a heart attack where the muscle is suddenly damaged.
(2) It is a state where the heart stops beating suddenly (cardiac arrest).
(3) It is the state of the heart when it is not pumping blood effectively enough to meet the body's needs.
(4) It is consistently high blood pressure above 140/90 mm Hg.

174. The innermost layer of the cranial meninges, which is in thin contact with the brain tissue, is the:

(1) Dura mater (2) Arachnoid mater
(3) Pia mater (4) White matter

175. Regarding the conducting part of the human respiratory system, which of the following is **not** a function it performs?

(1) Transportation of atmospheric air to the alveoli.
(2) Clearing the air from foreign particles and humidifying it.
(3) Bringing the inspired air to body temperature.
(4) Providing the site for actual diffusion of O_2 and CO_2 between blood and air.

176. Which of the following statements about the Thymus gland is **incorrect**?

(1) It plays a major role in the development of the immune system.
(2) It secretes peptide hormones called thymosins.
(3) Thymosins promote the differentiation of B-lymphocytes for cell-mediated immunity.
(4) The thymus is degenerated in old individuals, leading to a weakened immune response.

177. Match **List-I** with **List-II**.

List-I (Excretory structure)		**List-II (Representative organism)**	
(A)	Flame cells	(I)	*Locusta*
(B)	Nephridia	(II)	*Fasciola*
(C)	Malpighian tubules	(III)	Prawn
(D)	Antennal glands	(IV)	*Pheretima*

Choose the **correct** answer from the options given below:

(1) A-IV, B-III, C-I, D-II
(2) A-IV, B-II, C-I, D-III
(3) A-II, B-IV, C-III, D-I
(4) A-II, B-IV, C-I, D-III

178. A person performs a forced expiration. The maximum volume of air they can now breathe in is known as:

(1) Inspiratory Capacity (IC)
(2) Vital Capacity (VC)
(3) Total Lung Capacity (TLC)
(4) Functional Residual Capacity (FRC)

179. Which of the following is **true** regarding the mechanism of hormone action for steroid hormones?

(1) They generate secondary messengers like cAMP and IP_3.
(2) They interact with membrane-bound receptors to trigger a biochemical cascade.
(3) They enter the target cell and bind to intracellular receptors to regulate gene expression.
(4) They are short-lived and cannot cross the lipid bilayer of the cell membrane.

180. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Connective tissues are the most abundant and widely distributed tissues in the body of complex animals.

Reason R: Except for blood, the cells in all connective tissues secrete structural protein fibres like collagen or elastin.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both A and R are true, and R is the correct explanation of A.
(2) Both A and R are true, but R is not the correct explanation of A.
(3) A is true, but R is false.
(4) A is false, but R is true.

■■■

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